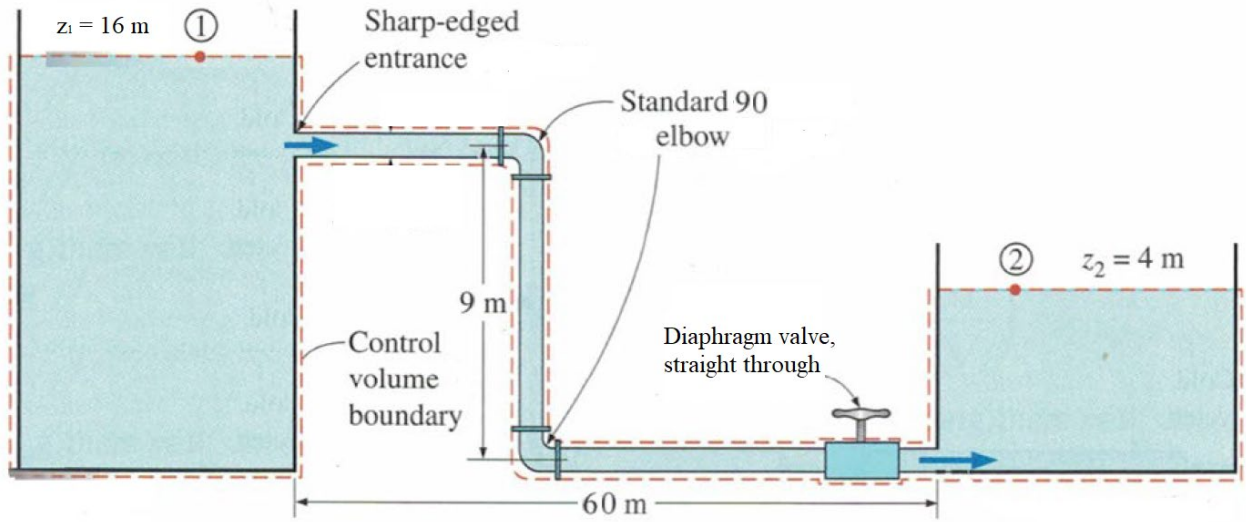
# ChE 2O04 Tutorial #8

**Tutorial Problem #1**

Your engineering consulting firm has been sent the design for a simple piping system connecting two large water reservoirs (both at 10 °C) as shown in the accompanying figure.




a. Build the system shown above in PIPE-FLO and report the volumetric flow rate of water (L/s) in the 150 mm (nominal size) cast iron Schedule 30 pipe when it is brand new and thus the corresponding roughness value is 0.25 mm (i.e. the smallest value in the Table on slide 46 of the PDF_07 notes).

*NOTE: You must provide a print-out of your FLO-SHEET (File → Print → FLO-Sheet Letter Landscape option so that it appears as a formal print out with the Program details, Units, and Project Information details displayed across the bottom). Simple screenshots are NOT acceptable.*

b. Manually calculate the volumetric flow rate of water (L/s) using the strategy outlined on slide 26 of the PDF_08 notes.

*NOTE: Your calculation should use the exact same pipe internal diameter and ∑K values from part a.*

c. Comment on the difference (if any) between the answers from parts a and b.

d. Your client has informed you that this system is scheduled to operate for many years and wants to know the effect of aging on the system performance. Fill in the table below by creating four new systems in PIPE-FLO identical to that described in part a:

| Pipe Roughness (mm) | Flow Rate of Water (L/s) |
|---|---|
| 0.25 | |
| 0.8* | |
| 1.5* | |
| 2.5* | |
| 5.0** | |

* taken from the Table on slide 46 of the PDF_07 notes

** based on the "twenty-fold increase" scenario (from Daugherty & Franzini's book *Fluid Mechanics with Engineering Applications*) which also appears on slide 46 of the PDF_07 notes.